

Predictive Power of Surprisal from LMs on Reading Times at Different Levels of English Proficiency

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Abstract

The study of eye movement in reading provides insights into language processing for both native (L1) and non-native (L2) speakers. Factors such as word length, frequency, and surprisal influence reading patterns in L1 readers and show a dynamic impact on L2 readers as their English proficiency changes. In particular, as the proficiency of L2 decreases, the eye movements of the readers diverge from L1 and become less sensitive to word predictability and more sensitive to word frequency (Berzak and Levy, 2023).

The ability of large language models to process language positions them as valuable tools in cognitive models of sentence processing. Counterintuitively to NLP paradigms of "Larger is Better", (Oh and Schuler, 2023) found that models with more parameters and lower perplexity often produce surprisal estimates that are less predictive of human reading times.

This study analyzes the effect of model perplexity in large transformer models and explores the impact of English proficiency on language processing. The results of the statistical analyzes support the conclusions of (Berzak and Levy, 2023) regarding the eye movements L1 and L2 speakers.¹

1 Introduction

Investigating the cognitive mechanisms that govern language comprehension is essential to advance our understanding of first-language (L1) and second-language (L2) acquisition. Prior research, including work by (Berzak and Levy, 2023), has explored the impact of linguistic factors such as surprisal, frequency, and word length on English language learners using the CELER dataset (Berzak et al., 2022). Their findings highlight both the similarities and differences in how native and non-native readers process language, suggesting that frequency

effects are linked to lexical retrieval while surprisal effects pertain to contextual integration.

While the findings of (Berzak et al., 2022) were based on surprisal extracted from the GPT2 model (Radford et al., 2019), this study aims to replicate these findings by employing a multitude of large language models (LLMs) of varying sizes. Specifically, Pythia models (Biderman et al., 2023) with 70M, 160M, and 410M parameters were used to calculate surprisal values.

Recent research by (Oh and Schuler, 2023) provides a vital context for this study. (Oh and Schuler, 2023) conducted a linguistic analysis to explore why larger Transformer-based models, despite having more parameters and lower perplexity, sometimes produce surprisal estimates that are less predictive of human reading times. Their work reveals a clear inverse relationship between model perplexity and the accuracy of reading time estimates on a native-language dataset alone (Barrett et al., 2015). This pattern was consistent in three different types of models, each tested with various sizes, replicating previous findings that were limited to GPT-2 (Oh et al., 2022).

These results provide empirical evidence supporting the notion that Transformer-based models tend to 'memorize' sequences during training, leading to surprisal estimates that deviate from human expectations. By replicating (Berzak and Levy, 2023)'s findings using different Pythia models with varying perplexities, this study aims to demonstrate a similar deviation from human L1 and L2 language processing.

The work will assess how model sizes influence surprisal measures and their predictive power for reading times across various levels of English proficiency. The outcomes aim to provide insights into how different LLMs capture surprisal effects and emphasize the considerations needed for selecting larger pre-trained language models to study language processing.

¹Code is available at: <https://github.com/NaomiDerel/multi-model-traces-of-ling-knowledge>

2 Data

The CELER (Corpus of Eye Movements in L1 and L2 English Reading) corpus (Berzak et al., 2022) comprises eye-tracking data collected from 365 participants, including 69 native English speakers (L1) and 296 English learners (L2) from five native language backgrounds (Arabic, Chinese, Japanese, Portuguese, and Spanish). The corpus contains over 320,000 words from 28,548 sentences selected from the Wall Street Journal (WSJ). Each participant read 156 newswire sentences, some of them shared among the participants and others unique to each participant. This design resulted in a comprehensive dataset combining individual and shared reading experiences.

The English proficiency of L2 participants in the CELER dataset was evaluated using the Michigan English Placement Test (MPT), whose scores range from 0 to 50, reflecting their accuracy in answering the 50 total questions. The scores were mapped to the Common European Framework of Reference for Languages levels as follows: 0–16 corresponds to A1 (beginner), 17–21 to A2 (elementary), 22–31 to B1 (intermediate), 32–36 to B2 (upper-intermediate), and 37–50 to C1 (advanced) (Council of Europe, 2001). This assessment provided a standardized measure of English proficiency, ensuring a diverse range of language abilities among the L2 participants, which is crucial for analyzing the impact of proficiency on reading behaviors.

To avoid correlations between samples and ensure robust analysis, this study focused on the Individual Regime of the CELER dataset. This regime includes 28,470 sentences (320,360 words), divided into 356 batches of 78 sentences each. The final dataset used in this paper contains 27,736 sentences and 14,561 unique words, after excluding out-of-vocabulary words, skipped words, punctuation, numbers, and sentence boundary words.

3 Experiments and Results

3.1 Terminology

The following presents the key terms and measures needed to understand the analysis of reading behavior and word properties.

Eye Movement Measures We focus on three standard fixation measures to assess reading behavior in our analysis:

1. First Fixation: the duration of the initial fixation on a word.

2. Gaze Duration: the total time spent from first entering the word until first leaving it.
3. Total Fixation: sum of all the fixations on a word.

These measures are related, each capturing progressively later stages of language processing. The First Fixation is the initial period within the Gaze Duration, and the Gaze Duration is included within the Total Fixation time. This combination of measures reflects the successive stages of processing a word, from initial recognition to more in-depth processing (Inhoff, 1984; Liversedge and Findlay, 2000; Rayner, 1998).

Word Properties The main properties we used are as mentioned: word length, word frequency, and surprisal.

The word frequency used is negative log-frequency, defined as $-\log_2 p(w)$, it is taken from the standard frequency list SUBTLEX-US (Van Heuven et al., 2014).

Surprisal is a common measure used in computational linguistics and psycholinguistics to quantify the unpredictability of a word in a given context. It is defined as $-\log_2 p(w|context)$, with the context being the rest of the sentence for $w \in sentence$.

GAM Models Similarly to (Berzak and Levy, 2023), we assume that the connection between word property and eye movement measure might not be linear. We employ Generalized Additive Models (GAMs) (Hastie, 2017), a flexible class of models that allow for non-linear relationships by fitting a sum of smooth functions.

Effect Size In order to measure the participant’s effect size for a GAM model, the analysis will include the average slowdown per participant. With notation from (Berzak and Levy, 2023), $s_{i,p}$ is the partial effect function, which might be non-linear, and C is the full corpus. Participant’s i slowdown on a word w via a word property $wp \in length, frequency, surprisal$ will be:

$$Slowdown_{i,p} = \frac{1}{|C|} \sum_{w \in C} s_{i,p}(wp(w)) \quad (1)$$

To calculate the significance of surprisal on reading time predictions following (Oh and Schuler, 2023). Using the GAM model, the log-likelihood is calculated separately for surprisal and for frequency, word length. Therefore, the measure for the significance of surprisal in the prediction is defined as the log-likelihood of surprisal exclusively.

3.2 Surprisal from LLMs

The Pythia models (Biderman et al., 2023) are a suite of large language models designed to facilitate research by providing a controlled, reproducible framework for studying the behavior and performance of various models. They offer 8 different models of increasing parameter sizes and intermediate checkpoints in training. This paper worked on Pythia-70M, 160M, 410M, the capacities of which are shown in Table 1. We leave further studies on larger models to future work, due to computational constraints.

Model	#L	#H	d	#Non-Emb Params
Pythia 70M	6	8	512	18.9M
Pythia 160M	12	12	768	85.1M
Pythia 410M	24	16	1024	302.3M

Table 1: Specifications of relevant Pythia models. #L, #H, d, and #Non-Emb Params refer to number of layers, number of attention heads, embedding size, and number of non-embedding parameters respectively.

Each model is used to calculate surprisal values, using the ability of LLMs to output $p(w|context)$, with the sentences in the CELER corpus inputted to the model consecutively, making the context of each word the full sentence before it.

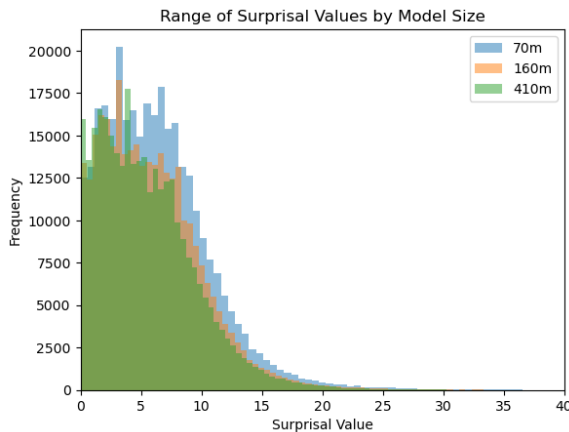


Figure 1: Distribution of Surprisal Values by Pythia Model Size, Showing Density Variations Across Different Parameter Configurations.

Trends in Surprisal Values by Model As illustrated in Figure 1, the distribution pattern of surprisal values remains consistent across different models. Additionally, it reinforces the notion that larger models tend to converge towards generally lower surprisal values, with each successive size increase resulting in a narrower range for the majority

Model	Mean	SD	Range
70M	6.341***	4.386	0.000 – 87.039
160M	5.884***	4.152	0.001 – 73.468
410M	5.518***	4.018	0.000 – 68.879

Table 2: Overview of surprisal values based on Pythia model sizes (rounded to three decimal places). Significantly different means for different models are indicated by $p < 0.001$.

of its values.

To evaluate the mean surprisal values among different Pythia models (70M, 160M, and 410M), an independent two-sample one-sided t-test on the means shown in Table 2 was conducted. The results indicate with statistical significance that larger models exhibit lower mean surprisal than their smaller predecessors.

Perplexity Each model also has a measure of perplexity, which is defined for a tokenized sequence $X = (x_0, \dots, x_t)$:

$$Perplexity(X) = \exp \left[-\frac{1}{t} \sum_i^t \log(p(x_i|x_{<i})) \right]$$

Where $\log(p(x_i|x_{<i}))$ is the log-likelihood of the i^{th} token conditioned on the preceding tokens $X_{<i}$.

The perplexity measure is calculated with code by (HuggingFace, 2020) on the Pythia models, using the WikiText-2 database (Merity et al., 2016). The results are shown in Table 3.

Model	Perplexity
70M	39.427
160M	23.303
410M	13.984
GPT2	16.45

Table 3: Perplexity of Pythia models and GPT2-Large on the WikiText-2 database.

3.3 Analysis 1: Functional Form

In Analysis 1, a Generalized Additive Model (GAM) was applied to explore the relationship between word properties and reading time, in L1 and L2 readers. Following (Berzak and Levy, 2023), the curves per word property are fitted using the Bam function (Wood et al., 2015). Surprisal curves

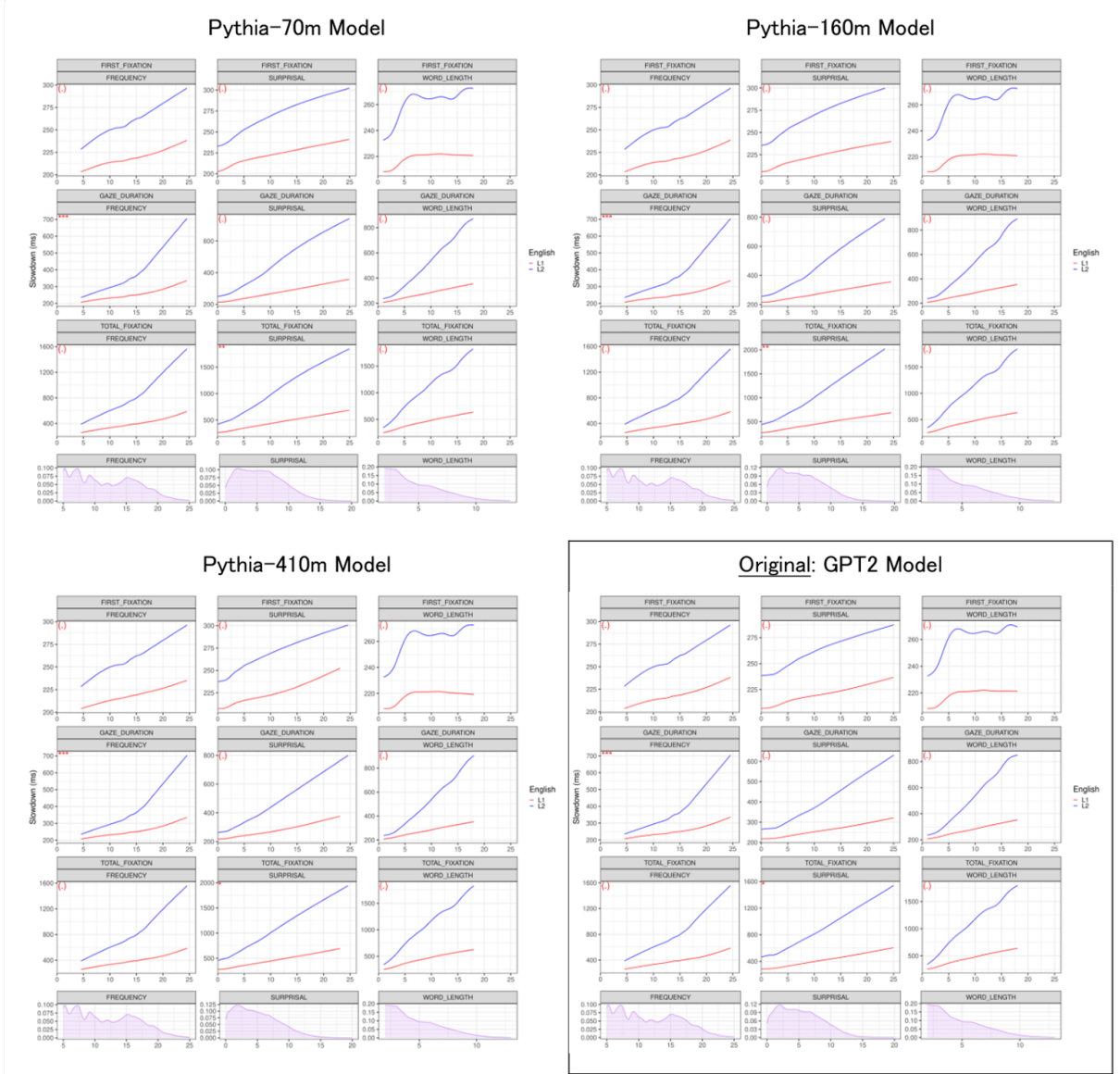


Figure 2: **GAM fits for the relation between benchmark word properties (current word) and raw reading times** by Equations 2 and 3, for each LLM. Per LLM, the upper rows depict the effect size (1) for eye movement measures as a function of word properties, and the bottom row shows the density plot for word properties.

were fitted with the model:

$$\begin{aligned}
 RT \sim & s(surp, bs = "cr", k = 20) \\
 & + s(surp_{pr}, bs = "cr", k = 20) \\
 & + te(freq, len, bs = "cr") \\
 & + te(freq_{pr}, len_{pr}, bs = "cr") \quad (2) \\
 & + s(subj, bs = "re") \\
 & + s(subj, surp, bs = "re") \\
 & + te(subj, freq, len, bs = "re")
 \end{aligned}$$

and the frequency and word length curves are

fitted with the model:

$$\begin{aligned}
 RT \sim & s(surp, bs = "cr", k = 20) \\
 & + s(surp_{pr}, bs = "cr", k = 20) \\
 & + s(freq, bs = "cr", k = 20) \\
 & + s(freq_{pr}, bs = "cr", k = 20) \\
 & + s(len, bs = "cr") \\
 & + s(len_{pr}, bs = "cr") \quad (3) \\
 & + s(subj, bs = "re") \\
 & + s(subj, surp, bs = "re") \\
 & + s(subj, freq, bs = "re") \\
 & + s(subj, len, bs = "re")
 \end{aligned}$$

Figure 2 presents the fitted models based on the

surprisal values from the relevant Pythia models, alongside the GPT-2 model used in the original study for comparison. Each plot illustrates the relationship between a predictor variable related to word properties and a response variable derived from eye movement measurements. The dual objective is to demonstrate the reproducibility of the original GPT-2 results with the smaller Pythia models, and to examine how model performance changes with increasing perplexity.

Method This analysis differs from (Berzak and Levy, 2023), where bootstrapping was used to fit the models. Here, the Generalized Additive Model (GAM) was fit directly to the entire dataset, foregoing bootstrapping due to time constraints. However, fitting the model to the full dataset introduced a challenge: the resulting plots, incorporating all data points, became cluttered and noisy, obscuring meaningful patterns.

To resolve this, the results were smoothed using the `geom_smooth` function to produce clearer visualizations of the underlying trends. Smoothing was necessary because the original, unsmoothed plots, which represented the full dataset, were difficult to interpret due to the high density of data points, resulting in visually noisy and cluttered graphs. The original unsmoothed plots can be found in Appendix A.

Results The key findings of (Berzak and Levy, 2023) were successfully replicated across different model sizes, highlighting the robustness of the original results. This replication confirms that their conclusions are consistent even when applied to models of varying sizes. The resulting curves were very similar across models, with the primary difference being the distribution of surprisal values rather than their impact on reading times.

It is important to recognize that not all surprisal values follow the expected linear trend observed in the analysis. This departure from linearity is particularly noticeable at higher surprisal values, which are considered outliers (A). A plausible explanation for this phenomenon is the limited number of words with high surprisal values in the dataset, as shown in Figure 1. The histogram indicates that words with high surprisal are relatively rare, resulting in a scarcity of data points in the upper percentiles of the surprisal distribution.

This under-representation of higher surprisal values reduces the statistical power to detect a clear linear relationship. In other words, with fewer in-

stances of high-surprisal words, the model may find it challenging to accurately capture and generalize the relationship between surprisal and reading time in this range. As a result, the effects of surprisal on reading time may appear less consistent at higher surprisal levels, leading to deviations from the expected linear trend.

Confidence Intervals The analysis initially used classical confidence intervals (CIs) calculated from the mean and standard deviation. However, the high density of the data resulted in very low standard deviation values, and the use of smoothed plots made it impractical to display accurate confidence intervals.

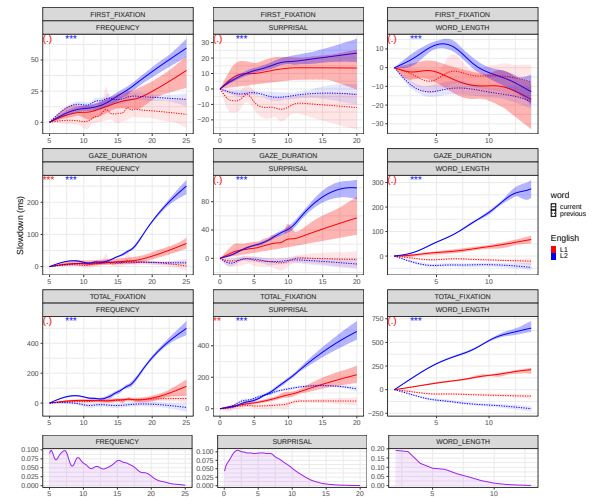


Figure 3: GAM fit for the relation between benchmark word properties (current & previous word) and raw reading times using the model. Estimated using Bootstrap with $B=13$

To address this limitation, bootstrap resampling was performed using surprisal values from the Pythia 70M model. This model was selected due to its smaller size and higher noise level, which would benefit most from confidence intervals, thereby improving the robustness of the analysis. Figure 3 shows the results obtained from the bootstrap procedure, confirming the successful replication of the original findings. The most significant difference from the results in (Berzak and Levy, 2023) appears in the plot relating eye movement metrics to surprisal, as this plot spans a wider range of values. We attribute this difference to the presence of more high-end surprisal values compared to GPT-2, along with fewer bootstrap iterations, which led to larger confidence intervals.

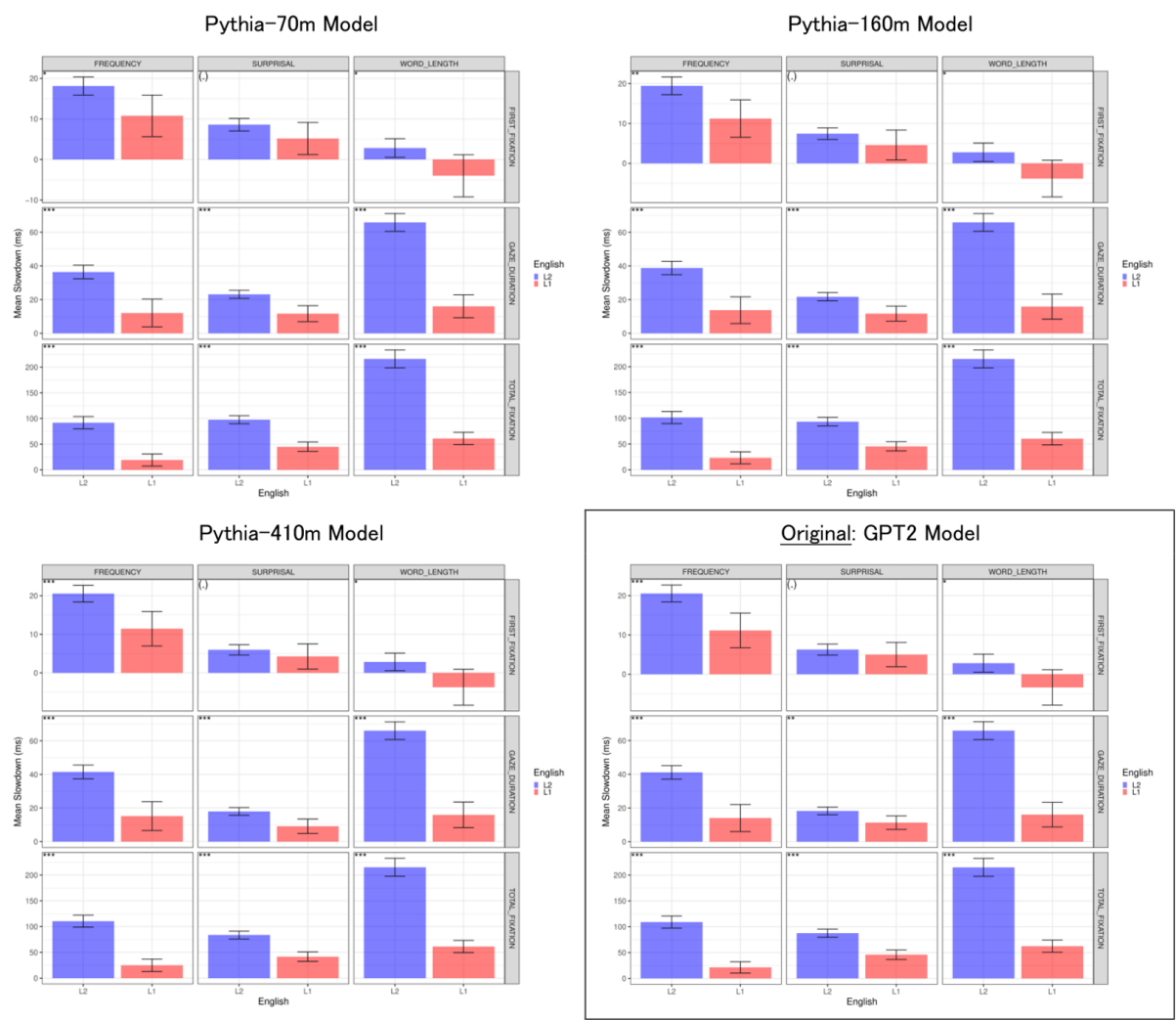


Figure 4: **Mean per subject slowdown effects in ms with 95% confidence intervals**, based on GAM model in Equation 2, 3 without subject random effects, fitted separately for each subject. Statistical significance of a t-test for the difference between L1, L2 is shown with ‘***’ $p < 0.001$, ‘**’ $p < 0.01$, ‘*’ $p < 0.05$, ‘(.)’ $p > 0.05$.

3.4 Analysis 2: Magnitude

In Analysis 2, (Berzak and Levy, 2023) summarized the findings from Analysis 1 to quantify and compare the overall effects of surprisal in both first language (L1) and second language (L2) reading. Earlier research by (Whitford and Titone, 2017) found no significant differences between L1 and L2 for predictability effects. Contradicting this, (Berzak and Levy, 2023) further explored the dynamics of surprisal and frequency and found that surprisal has a greater impact on Gaze Duration and Total Fixation in L2 compared to L1. This analysis was done identically and corroborates these findings by assessing how surprisal influences reading times across models of varying sizes. The results are consistent with those of (Berzak and Levy, 2023), underscoring the robustness of the effects

across different model sizes. The only nonsignificant result is the surprisal effect on first fixation for both L1 and L2 readers.

3.5 Analysis 3: Interaction with L2 Proficiency

In this key analysis, performed identically to (Berzak and Levy, 2023), the focus is on how responsiveness to word properties changes with English L2 proficiency. This follows a similar approach to Analysis 2, where a GAM is fitted for each participant. The effect size is measured using Equation 1, as a function of English proficiency, measured by the listening comprehension and grammar sections of the Michigan Placement Test (MPT).

Figure 5 illustrates the resulting curves compared to the average slowdown of the L1 group.

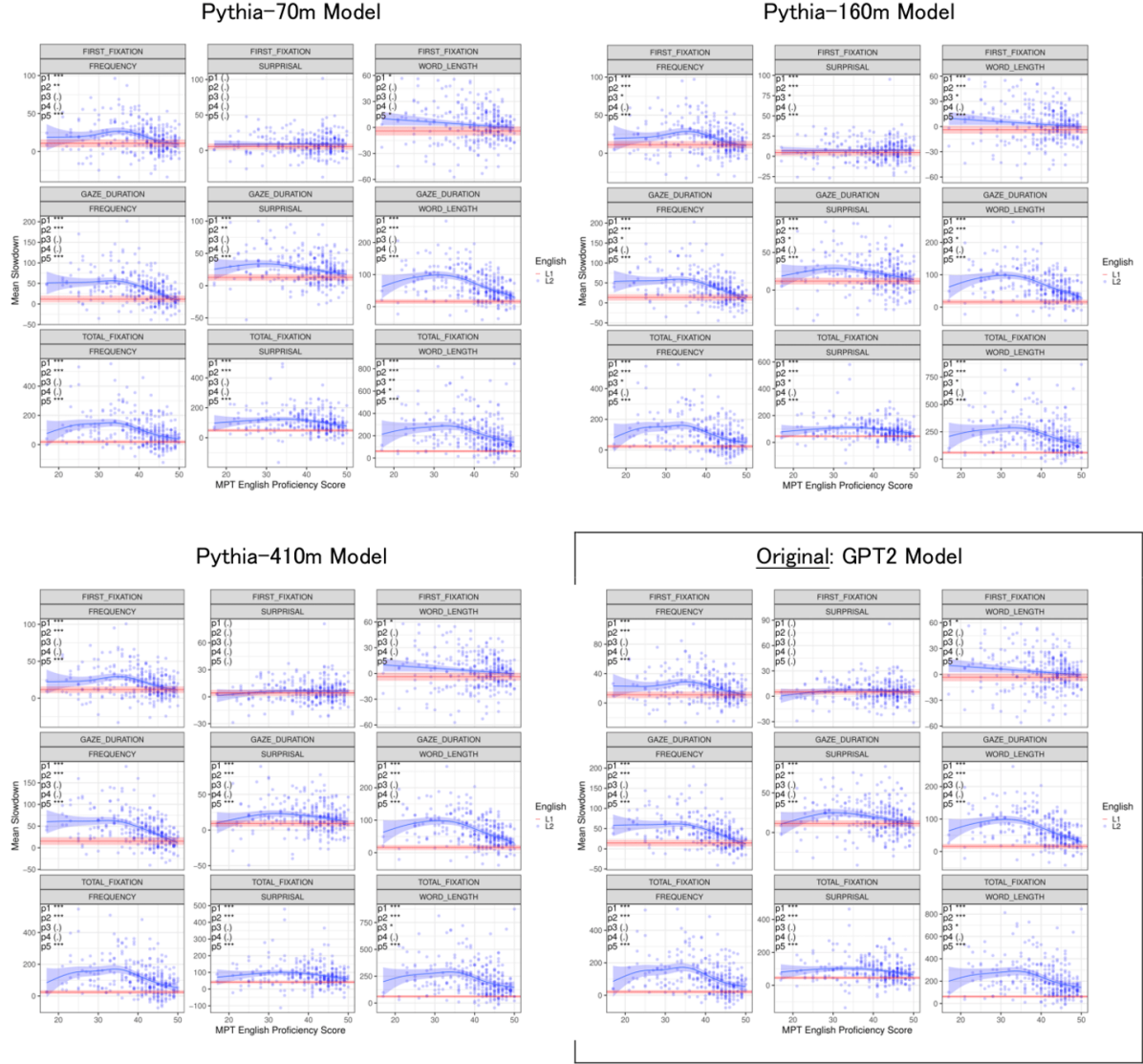


Figure 5: Slowdown effects associated with benchmark word properties of the current word for raw reading times as a function of English proficiency. With GAM model from 2, 3, fitted on each participant and measure separately. Slowdown (1) is described as a function of the MPT English proficiency score. The red line shows mean of L1 slowdown effect, while the blue line is the GAM fit for L2 speakers as represented by the dots.

The results were replicated across all three Pythia models. The results show a statistically significant U-shaped relation between proficiency and frequency across all three fixation measures, as well as surprisal for Gaze Duration and Total Fixation.

Pythia 70M shows the most noticeable deviation, presenting a slight departure from the previously observed U-shape pattern. Nonetheless, even this model displays results that are consistent with the original behavior fitted to GPT-2 surprisal.

The Relationship Between Proficiency and Response to Word Properties Five statistical tests are conducted in 5 to answer questions about the

relationship between English proficiency level, and the effect of word properties as defined previously. The results correspond to the numbered p-values in this Figure.

$$m1 : response \sim s(MPT)$$

$$m2 : response \sim MPT + MPT^2$$

$$m3 : response \sim English + MPT$$

$$m4 : response \sim English + MPT + MPT^2$$

$$m5 : response \sim MPT$$

The first 2 tests address the non-linearity of the effects of L2 English proficiency, which are signifi-

cant in surprisal for gaze duration and total fixation, across all models. Tests 3 and 4 compare the most proficient L2 speakers to L1 speakers, calculated following the methodology in (Berzak et al., 2022), and test 5 is applied only to L2 readers with above-average scores, and compares the effect size about proficiency. These findings support the conclusion drawn from the plots that the most fluent L2 speakers are similar to L1 speakers.

3.6 Effect of Model Size on Predictions

Model	L1 Log-Likelihood	L2 Log-Likelihood
70M	-185,516	-1,033,348
160M	-185,512	-1,033,355
410M	-185,514	-1,033,411

Table 4: Log-Likelihood of Pythia models for GAM Surprisal on Gaze Duration.

This work aimed to compare the effect of model size, or inversely model perplexity, on the predictive power of surprisal on reading times. Unexpectedly, previous analysis and results showed incredibly similar results among all models, from much smaller Pythia-70m to larger GPT2. To further compare the models, the log-likelihood of the surprisal GAM model described in Equation 2 was calculated for each model. The results, as shown in Table 4 for the Gaze Duration measure, show that the model size does not affect the Log-Likelihood. This holds for all measurements of eye movement identically (detailed in Appendix A).

Notably, the log-likelihood for L1 readers is much higher than L2 readers, which indicates a better fit to the data. This supports other analyses in the conclusion that surprisal is more significant in predicting reading times for L1 readers over L2 readers.

4 Discussion and Conclusions

In the preceding sections, we replicated the findings from (Berzak and Levy, 2023) using models of varying sizes, specifically those with 70M, 160M, and 410M parameters. We extended the original study’s analyses, which examined the relationship between reading times and word properties, effect sizes, and responsiveness to L2 English proficiency. We also compared log-likelihood and surprisal distributions across these models.

Our analyses closely matched the original experiment, supporting two key claims. First, (Berzak

and Levy, 2023) suggested that surprisal and frequency affect reading differently, with frequency influencing lexical processing and surprisal affecting contextual processing. Consistent results across different models reinforce the importance of surprisal in understanding reading processes. Second, (Berzak and Levy, 2023) argued that language model-based surprisal is less accurate in estimating L2 probabilities than L1. This pattern persisted even with smaller models and in the last experiment, indicating that challenges in modeling L2 reading behavior are not solely due to model size but may stem from deeper differences in L1 and L2 processing.

These findings align with (Oh and Schuler, 2023)’s suggestion that smaller models can effectively capture the impact of surprisal on reading time. However, contrary to expectations, our experiments consistently produced results similar to those of (Berzak and Levy, 2023), regardless of model size. The anticipated conclusion that smaller models would better fit human reading times was not supported in this work.

Our main contribution lies in demonstrating the robustness of these findings across a range of model sizes, underscoring the reliability of surprisal as a key factor in reading time analysis. Our work also supports valuable and actionable insights in the field of cognition and language processing, particularly in the infrequently studied domain of native and non-native speakers.

4.1 Future Work

Future research could broaden the scope by incorporating additional models, overcoming the current study’s time and computational constraints. Larger models could be explored to investigate the potential decline in performance with increasing size, while smaller models could be examined to identify peak performance. Alternatively, this notion might be disproved entirely.

Furthermore, applying bootstrapping techniques to all models and generating confidence intervals is a crucial task. Although this aspect of the analysis could not be completed due to equipment and time limitations, it is vital for enhancing the robustness and reliability of the results and reaching definitive conclusions.

Addressing these points in future work will contribute to a more comprehensive understanding and validation of the study’s findings.

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A Appendix

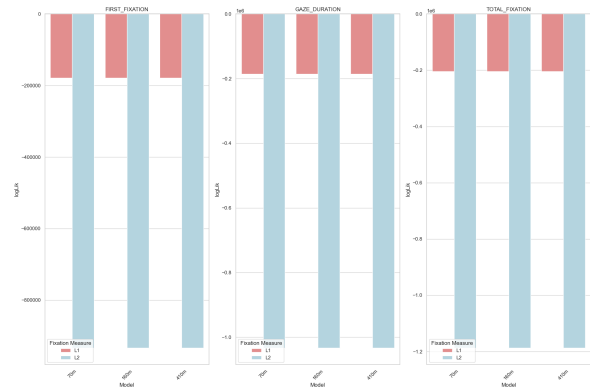


Figure 6: Log-Likelihood of Pythia-70, Pythia-160, and Pythia-410 over L1 and L2

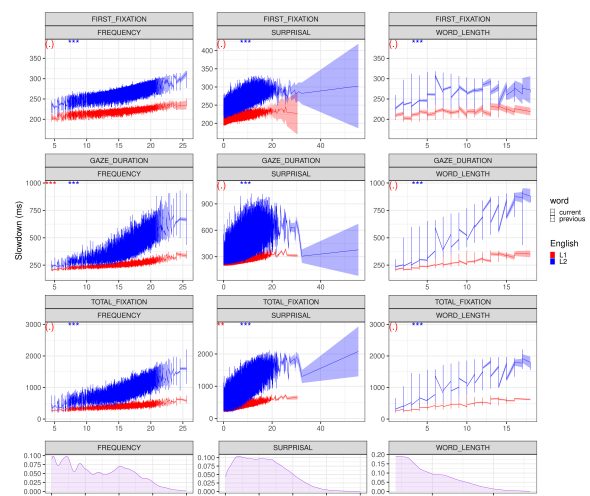


Figure 7: Unsmoothed Analysis 1 - Pythia 70m model: GAM fits for the relationship between word properties and raw reading times for the full dataset, without smoothing.

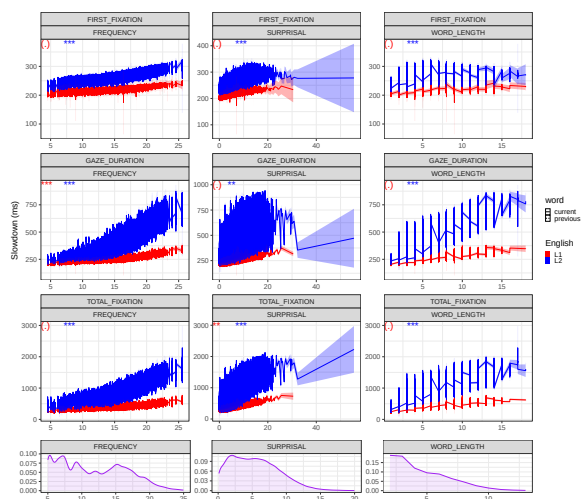


Figure 8: **Unsmoothed Analysis 1 - Pythia 160m model:** GAM fits for the relationship between word properties and raw reading times for the full dataset, without smoothing.

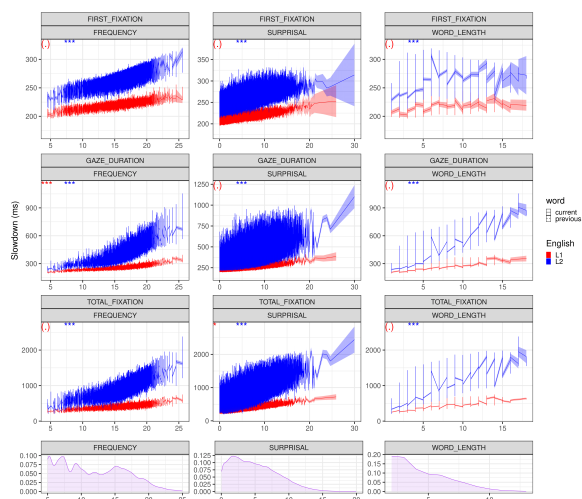


Figure 9: **Unsmoothed Analysis 1 - Pythia 410m model:** GAM fits for the relationship between word properties and raw reading times for the full dataset, without smoothing.